

Lab01_deBarros

Matheus de Barros

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The question:

How does the length of the training data affects resulting forecasts from ARIMA models?

The methods

Different ARIMA models were fitted to multiple sets of the monthly data on chinook landings split into varying intervals, namely 5, 10, 15, 20, and 25 years of data. Then, forecasts based on each of the training datasets were created and compared with the observed values from the raw dataset.

Formatting the data

```
chinook.month = chinook.month %>%  
  mutate(Date = as.Date(paste(Year, Month, "01", sep = "-"),  
    format = "%Y-%b-%d"))  
  
chinook.month$Month2 = as.numeric(as.factor(chinook.month$Month))
```

Splitting the data into different sets of training and testing data with varying lengths

```
month_ts = ts(chinook.month$log.metric.tons, start = c(1980, 1), frequency = 12)  
  
# Create test data with different ts lengths  
ts_lengths = c(5,10,15,20,25)  
month_train = list()  
month_test = list()  
for (i in 1:length(ts_lengths)) {  
  
  month_train[[i]] = window(month_ts,  
    start = c(1980, 1),  
    end = c(1980 + ts_lengths[i], 12))  
  
  month_test[[i]] = window(month_ts,  
    start = c(1980 + ts_lengths[i] + 1, 1),  
    end = c(2019, 12))  
  
}
```

Fitting models using ‘auto.arima’

```
# Fit ARIMA models to all time series  
train_models = list()
```

```

for (i in 1:length(month_train)) train_models[[i]] = auto.arima(month_train[[i]])

# Compare forecast against test data
fr = list()
for (i in 1:length(month_train)) fr[[i]] = forecast(train_models[[i]], h = length(month_test[[i]])*2)

plots = list()
for (i in 1:length(month_train)) {
  plots[[i]] = autoplot(fr[[i]]) +
    geom_point(aes(x = x, y = y), size = 0.5, data = fortify(month_test[[i]])) +
    labs(title = paste("Number of years =", ts_lengths[i]), x = "Year", y = "Log(mt)") +
    theme_minimal()
}

```

Discussion

As expected, increasing the number of years on the training data significantly impacts resulting forecasts such that they become more accurate and precise (less uncertain). These improvements are likely due to (1) improve parameter estimation, (2) improved detection of long-term patterns such as seasonality, and (3) decreasing influence of outliers as the training set grows.

```
ggarrange(plotlist = plots, ncol = 2, nrow = 3)
```

```

## Warning: Removed 47 rows containing missing values or values outside the scale range
## (`geom_point()`).

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## (`geom_point()`).

```

